

Coordinated operational optimization of water and power systems under emergency conditions

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ABSTRACT

This research addresses the integrated management of Water Distribution Systems (WDS) and Power Distribution Systems (PDS) to enhance operational efficiency and resilience during extreme scenarios. Traditionally, these systems are operated independently, leading to sub-optimal performance. Alternatively, some studies have proposed a joint operation of WDS and PDS, assuming that decisions are made simultaneously by a single decision-maker, which is impractical. We propose a novel emergency control method that relies on sparse communication between WDS and PDS operators, aiming to minimize load shedding (LS) in the PDS by strategically managing power demand in the interconnected WDS. The results show that the proposed coordinated approach has similar performance to a full cooperation approach between the two systems while requiring only minimal information exchange. Our approach demonstrates the potential of limited yet targeted information exchange, fostering cross-sectoral collaboration to maintain service continuity under extreme conditions, and underscores the critical role of integrated management in ensuring the resilience of interconnected infrastructure systems.

1. Introduction

With the increasing demands on Power Distribution Systems (PDS) and the challenges posed by integrating renewable energy sources, ensuring system resilience and efficiency through smart grid management has become a critical objective. The intermittent and uncertain nature of renewable sources increases the vulnerability of PDSs to extreme events, hazards, and instabilities, particularly due to rapid changes in weather conditions. Therefore, developing robust strategies to maintain PDS functionality across a broad spectrum of situations is crucial. Nevertheless, the current trends in power supply management pose substantial challenges in achieving this goal. Integration of renewable energy sources and energy storage facilities impacts network topology. Networks become more decentralized, thereby facilitating better management of energy resources at the local level to improve

overall grid efficiency. As a result, portions of the power network function as off-grids microgrids that can operate independently of the main transmission grid [1]. While off-grid microgrids are robust to failures in the transmission grid, their isolation can exacerbate the challenges of maintaining system stability and meeting demand, especially under extreme conditions [2].

Extensive research has been conducted on enhancing PDS resilience against such scenarios, including studies from a preventive perspective such as system design [3–5] as well as corrective aspects focusing on mitigating the extent and severity of extreme events [6–8]. A common strategy for reducing the impact of these events is load shedding (LS) which becomes necessary when the system's capacity to meet demands is compromised, leading to enforced power outages. LS is defined as power demand not supplied [9]. Under such circumstances, PDS operators are tasked with making operational decisions to minimize the impact on consumers in terms of LS [10–12]. A promising strategy for LS

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Nomenclature*Notation meaning*

$LS_{i,t}$	Load Shedding at bus i at time t
CI_b	Criticality Index for bus b
$\mathcal{B}$	Set of all buses
$\mathcal{L}$	Set of all power lines
$\mathcal{G}$	Set of all generators
$\mathcal{E}$	Set of all batteries
$P_{g,t}$	Active power generated by generator g at time t
$Q_{g,t}$	Reactive power generated by generator g at time t
$P_{i,t}^L$	Active power load at bus i at time t
$Q_{i,t}^L$	Reactive power load at bus i at time t
$P_{i,t}^P$	Active power load for pumping at bus i at time t
$SOC_{e,t}$	State of charge of battery e at time t
$\bar{P}_{g,t}, \underline{P}_{g,t}$	Min and max generation bounds for generator g at time t
$P_{ch,e,t}$	Battery e charging\discharging at time t
$\bar{SOC}_{e,t}, \underline{SOC}_{e,t}$	Min and max energy storage capacity for battery e at time t

$\bar{P}_{ch,e,t}, \underline{P}_{ch,e,t}$	Min and max battery charging rates
$r_{j,i}$	Resistance of line from bus j to bus i
$x_{j,i}$	Reactance of line from bus j to bus i
$I_{(j,i),t}^{sq}$	Squared current from bus j to bus i
$V_{i,t}^{sq}$	Squared voltage at bus i at time t
$\underline{V}_{i,t}^{sq}, \bar{V}_{i,t}^{sq}$	Min and max squared voltage bounds
$p \in \mathcal{P}$	Pump station (p) and the set of all pump stations ($\mathcal{P}$)
$s \in \mathcal{\Phi}$	Storage tank (s) and the set of all storage tanks ($\mathcal{\Phi}$)
$vol_{s,t}$	Tank s volume at time t
$\underline{vol}_{s,t}, \bar{vol}_{s,t}$	Tanks min and max volume
$x_{t,p,c}$	1 if pump station p is active with pumps combination c, 0 otherwise
$q_{p,c}$	Nominal flow of pump station p with units combination c
$P_{p,c}$	Nominal Power of pump station p with units combination c
$d_{s,t}$	Water demand from tank s at time t
γ_t	Electricity tariffs
ω	Pumping penalties

minimization is Demand Response (DR), where flexible loads are shifted to periods of lower demand to support the balance between supply and demand. In this context, infrastructure systems that are interconnected to PDS have the potential to serve DR, such as heating systems [13,14], gas distribution systems [15], and Water Distribution Systems (WDS) [16]. Recently there has been a growing interest in decision-making processes that integrate PDS and WDS. WDS are infrastructure in charge of the water supply for domestic, agricultural, and industrial uses and typically include power intensive facilities (pumping stations, desalination plants, etc.). As a result, WDS are among the largest power consumers accounting for an estimated 7–8 % of the total globally generated energy [17,18]. Moreover, WDS consists of inherent operational flexibility due to storage facilities that allow for operational adjustments and are typically used for cost minimization [19,20]. A set of power-consuming water facilities controlled by the same operator can offer significant potential to provide the required flexibility to both LS reduction and support many other PDS applications [21]. At the same time, several studies suggested joint optimization for WDS and PDS to minimize operational costs [22–25] and investment costs [26,27]. Furthermore, the same WDS flexibility can serve other DR objectives. For instance, [16] demonstrated how DR involving WDS can reduce GHG emissions while not compromising on operational costs. Additional applications of integrating WDS with PDS are frequency regulation [28] and voltage regulation [29]. The flexibility inherent in WDS can also be utilized to increase renewable energy source integration [30], acting as flexible loads for PDS that can absorb excess power or adjust to intermittent supply. The renewable surplus can also be effectively used by storing energy indirectly through pumping to fill water tanks during off-peak periods [31].

Due to the growing stress on both systems, interdependent resilience is identified as a crucial aspect of PDS and WDS management where the potential benefit is yet to be fully realized [32,33]. According to [34], the resilience dependency between the systems is bi-directional where each system can contribute to the other system's resilience. [34] also pointed out the challenges associated with integrated systems resilience enhancement and suggested a method to quantify the combined resilience. The quantification of multi-system resilience was the focus of several studies that suggested different metrics for this purpose. A common approach for quantifying resilience is according to the amount of lost power and water [35]. A different approach, suggested by [36], is to include metrics based on the quality of the power flow on top of the loss of service. Power quality is estimated as voltage magnitude metric,

duration of power lines at their upper bounds, and thermoelectric metric specific for the case where the WDS is used as a cooling cycle system. A few studies suggested methods for resilience enhancement through design and operational interventions. [37] presented a multi-objective optimization model to minimize investment costs and expected loss of power and water. A similar approach from an operational perspective was presented in [38]. An emergency response is suggested to a PDS that connects several microgrids. Once an emergency occurs, each microgrid adjusts its operation to support the entire distribution system by minimizing the total loss of power and water. Yang & Lo [39] introduced a multi-objective approach aimed at simultaneously minimizing unsupplied loads and water, the tradeoff between the two objectives is discussed presenting that loss of power can be prioritized over loss of water and vice versa.

The above studies presented fully collaborative decision-making frameworks, meaning that decisions for both systems are made simultaneously by a single decision maker. A contrasting perspective involves each utility making its own decision in response to the other utility's decision, without any cooperation. For instance, [35] presented an independent decision-making approach where the PDS operator determines an emergency response policy to minimize LS, and then the WDS operator adjusts pump scheduling according to the available power to minimize water supply shortage. [40] presented a non-cooperative game theory approach in which the two utilities act as independent players in a Cournot competition to determine the optimal operation dispatch. While the above studies present remarkable advancements in utilizing the links between the two systems, there still exists a significant knowledge gap: Cooperation-based approaches embody an ideal situation of complete knowledge sharing and cooperation, yet this is rarely the case in real life and is thus impractical. Conversely, independent approaches lack the benefits of collaboration and are therefore likely to be sub-optimal. Consequently, a strategy that enables collaboration through reasonable communication between PDS and WDS is essential for achieving more resilient and efficient infrastructure systems.

1.1. Contribution

This study focuses on improving PDS resilience, i.e., minimizing the impact of extreme scenarios on power consumers, by utilizing the operational flexibility of WDS. Specifically, we propose a collaborative approach to minimize LS under emergency scenarios with minimal information exchange between PDS and WDS. A key contribution of this

paper is the design of a modular optimization framework that optimizes PDS and WDS either individually or combined, utilizing WDS flexibility to minimize overall operational costs during routine operations or reduce LS under extreme conditions. Additionally, this study develops the paradigm of coordinated optimization, introducing an optimization strategy based on a practical communication protocol between PDS and WDS operators. This protocol enables system collaboration with only sparse communication yet achieves performance nearly equivalent to scenarios with full transparency and cooperation. The performance evaluation is demonstrated by comparisons with the two other common optimization approaches, namely independent and cooperation-based, and showcased through a case study of a local off-grid microgrid PDS connected to a local WDS across a range of extreme scenarios. The study also conducts an extensive analysis of the dependencies between properties of extreme scenarios and their impact on the LS reduction potential of various optimization strategies.

2. Methodology

2.1. Frameworks for integrated systems decision-making

To provide clarity and consistency throughout this paper, let us introduce several terms that will be used throughout this paper to distinguish between the different optimization decision-making strategies:

- **Decoupled Optimization:** In this approach, there is no communication between the two systems' operators. One operator takes action independently, and the second operator responds to these actions subsequently, without prior coordination.
- **Centralized Coupled Optimization:** This approach involves a fully cooperative strategy where decisions for both PDS and WDS are made simultaneously by a single stakeholder, ensuring that all objectives and constraints regarding both systems are known and considered in tandem.
- **Coordinated Optimization:** This new approach, introduced in the current study, is based on sparse communication and information exchange between the operators of PDS and WDS. It seeks to balance the benefits of Centralized Coupled optimization, which represents the optimal solution with the practicality of Decoupled optimization, which is simple and does not depend on any collaboration, aiming to

optimize the resilience of both systems through selective information sharing.

Fig. 1 depicts the overview of the proposed methodology. Emergency events are generated randomly based on several properties that describe the two systems' states (see a detailed description of the random scenarios generation in the last part of the methodology section). Each scenario is optimized with all the above decision-making strategies to find optimal operation policies, minimizing the loads not served.

The method is designed to minimize LS in PDS under extreme scenarios with limited power resources. The method is designed for a local microgrid is a medium voltage of 13.3 kV, but similar principles can be applied to other types of networks. The first step is to develop an optimization model with an objective function subject to physical and operational constraints. The optimization modeling allows for either a combined optimization of the WDS and PDS or separate optimization problems for each system. While the PDS is optimized to reduce LS, the WDS has two distinct objective functions. The first is for routine periods where the objective is to minimize operational costs. The second objective is for emergency scenarios where the WDS prioritizes supporting the PDS. In the second case, the WDS minimizes weighted pumping power consumption, where the weights are received from the PDS operators to represent the time steps and locations (buses) where LS reduction is most essential. The LS is calculated using a soft constraint approach involving adding slack terms to the bus power balance constraints which permits some degree of constraints violation. The sum of all violations is the PDS objective function. In the WDS problem, all constraints are hard to ensure that water supply is not compromised to mitigate power LS. Following are the mathematical formulations for the three frameworks of integrated decision-making. The notation used in the equations below is summarized in the Nomenclature table above.

2.2. PDS model

As mentioned above, the PDS objective is to minimize LS. To account for the differential criticality between buses, a criticality index (CI) is assigned to each bus to represent the importance of LS at that bus. The decision variables associated with the PDS are the active power generated in each generator ($P_{g,t}$) and the charging and discharging ($P_{ch,e,t}$) of each battery at every time step. The objective is to minimize the weighted LS calculated as the product of unsupplied demands (LS) and the CI

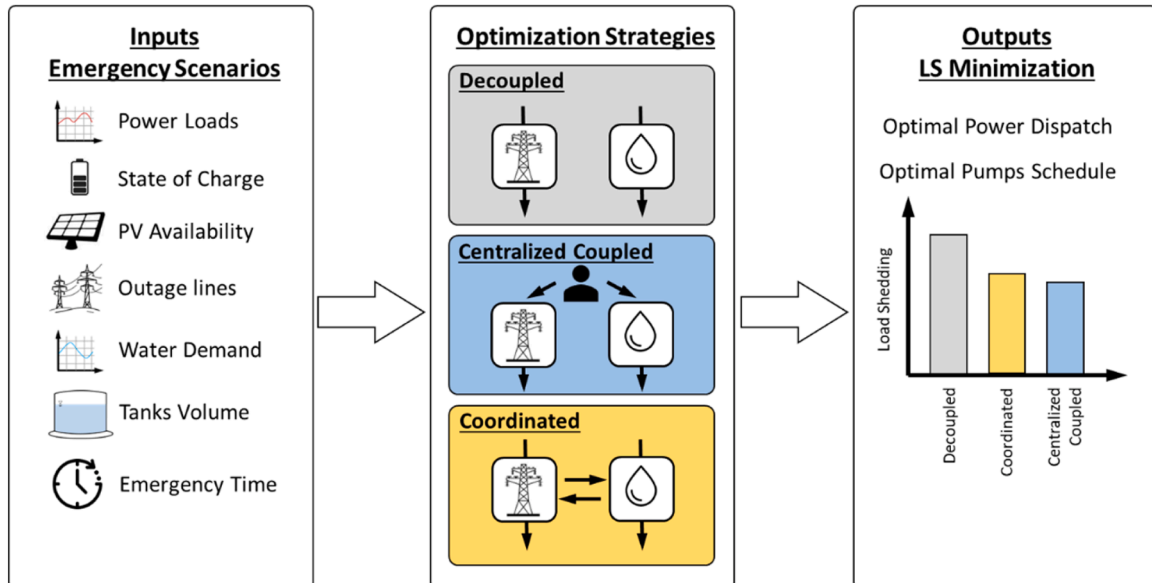


Fig. 1. Methodology Overview, LS minimization through different levels of PDS – WDS collaboration.

across all buses, see Eq. (1):

$$Z_{PDS \text{ resilience}} = \min_{P_{g,t}, P_{ch,e,t}} \sum_{i \in \mathcal{B}} \sum_{t \in T} LS_{i,t} \bullet CI_{i,t} \quad (1)$$

The PDS optimization is subject to several constraints. First, each generator has minimum and maximum capacity limits.

$$\underline{P}_{g,t} \leq P_{g,t} \leq \bar{P}_{g,t} \quad \forall t \in T, \quad \forall g \in G \quad (2)$$

While for conventional fuel-based generators, we assume these limits are constant over time, we model the variability of renewable generators (e.g., solar and wind) according to capacity depending on natural dynamics.

Batteries State of Charge (SOC) must satisfy energy conservation (3) and is bounded between the minimum and maximum capacities (4). Moreover, charging rates are also bounded between minimum and maximum power flow bounds (5).

$$SOC_{e,t} = SOC_{e,t-1} + P_{ch,e,t} \quad \forall t \in T, \quad \forall e \in E \quad (3)$$

$$\underline{SOC}_{e,t} \leq SOC_{e,t} \leq \overline{SOC}_{e,t} \quad \forall t \in T, \quad \forall e \in E \quad (4)$$

$$\bar{P}_{ch,e,t} \leq P_{ch,e,t} \leq \underline{P}_{ch,e,t} \quad \forall t \in T, \quad \forall e \in E \quad (5)$$

The active and reactive power balances are maintained at each bus for every time step. The active power balance is described in Eq. (6) and includes a penalty term for not supplying all the demand i.e., LS. It is noted that the pumps power term $P_{i,t}^p$ is used to couple between the PDS and WDS as later explained and presented in Eq. (18). The reactive power constraint is presented in Eq. (7). The unsupplied reactive power is neglected. Pumps are assumed to have a real-to-reactive ratio of 0.75, meaning a power factor of 0.8.

$$\sum_{g \in \mathcal{I}} P_{g,t} - \sum_{b \in \mathcal{I}} P_{ch,e,t} + \left(\sum_{g \in \mathcal{I}} P_{(j,i),t} - r_{j,i} I_{(j,i),t}^{sq} \right) - P_{i,t}^L - \sum_{p \in \mathcal{I}} P_{i,t}^p + LS_{i,t} = 0 \quad \forall t \in T, \quad \forall i \in \mathcal{B} \quad (6)$$

$$\sum_{g \in \mathcal{I}} Q_{g,t} + \left(\sum_{g \in \mathcal{I}} Q_{(j,i),t} - x_{j,i} I_{(j,i),t}^{sq} \right) - Q_{i,t}^L - 0.75 \sum_{p \in \mathcal{I}} P_{i,t}^p = 0 \quad \forall t \in T, \quad \forall i \in \mathcal{B} \quad (7)$$

Bus voltages must remain within specified minimum and maximum bounds.

$$\underline{V}_{i,t}^{sq} \leq V_{i,t}^{sq} \leq \bar{V}_{i,t}^{sq} \quad \forall t \in T, \quad \forall i \in \mathcal{B} \quad (8)$$

The voltage drops along lines is formulated in Eq. (9):

$$V_{i,t}^{sq} - V_{j,t}^{sq} = -2(r_{j,i} P_{(j,i),t} + x_{j,i} Q_{(j,i),t}) + (r_{j,i}^2 + x_{j,i}^2) I_{(j,i),t}^{sq} \quad \forall t \in T, \quad \forall j, i \in \mathcal{L} \quad (9)$$

And the power flow along lines is approximated to a rotated cone constraint based on Jabr inequality [41] which enables to maintain the problem's convexity.

$$V_{i,t}^{sq} I_{(j,i),t}^{sq} \geq P_{(j,i),t}^2 + Q_{(j,i),t}^2 \quad \forall t \in T, \quad \forall j, i \in \mathcal{L} \quad (10)$$

2.3. WDS model

The WDS optimization is framed as a pump scheduling problem subject to satisfying consumer demands. The hydraulic behavior of the WDS, including flow-head relationships, is bypassed using an implicit hydraulic formulation. This approach avoids the need to solve nonlinear hydraulic equations during optimization by relying on pre-computed system parameters and operational constraints that indirectly capture the connectivity and performance of the network. Such simplification is commonly employed in WDS optimization studies [42,43] to enhance

computational efficiency and scalability. The system operation is discretized to a finite number of hydraulic states, each representing a different set of active pumps. Table 1 presents this setup, where every row represents a set of active pumps such that a value of 1 indicates the pump is active (and 0 otherwise). The decision variables in this formulation are binary variables that determine whether a pump combination is used or not. Each active pump is associated with nominal flow and power values which are the results of the hydraulic conditions generated by activating the corresponding set of pumps.

To integrate the operation of the WDS operation with the PDS, an auxiliary matrix is introduced. This matrix (noted A below) represents the links between the WDS pumps and the PDS buses, with rows representing buses and columns representing pumps. The matrix elements i, j are set to 1 if pump j is connected to bus i and 0 otherwise. As mentioned above, two objective functions can be considered. The first is minimizing operational costs, calculated as the product of the pumping power and electricity tariffs (11).

$$Z_{WDS \text{ cost}} = \min_{x_{t,p,c}} \sum_{t \in T} \sum_{p \in \mathcal{P}} \sum_{c \in \mathcal{C}} \gamma_t x_{t,p,c} P_{p,c} \quad (11)$$

where γ_t is the electricity tariff at time t , $x_{t,p,c}$ is the binary decision variable equals 1 if combination c of pump station p is active at time step t . Lastly, $P_{p,c}$ is the nominal power consumed by pump combination c of pump station p .

A second objective function can be activated when PDS operators signal for an emergency. Given the signal, the water utility shifts the focus from operational costs to minimizing weighted energy consumption. Weights are provided by the power utility as a 2D matrix with rows corresponding to buses and columns to time steps. Each element $\omega_{i,j}$ in the matrix represents the prioritization of reducing pumping based on power supplied from bus i , at time step j . A higher value of $\omega_{i,j}$ indicates a stronger preference for pumping reduction.

$$Z_{WDS \text{ resilience}} = \min_{x_{t,p,c}} \omega \bullet A \bullet B \bullet x_{t,p,c} \quad (12)$$

Where ω is the weights matrix delivered by the power utility, A is the bus-pump connectivity matrix and B is the matrix that holds the pumping power for each combination of pumps (illustrated in Table 1), in other words, B holds all the $P_{p,c}$ values that are used in Eq. (11). This framework applies penalties based on the criticality of reducing pumping at certain times and locations.

The WDS optimization is constrained to satisfy mass balance (13), tank volume bounds (14), and tank final volume (15). Another constraint forces the solution to include only one combination of pumps for each pump station for each time step (16). Finally, the activation of a combination is a binary variable.

$$\sum_{p \in \mathcal{S}_{in}} \sum_{c \in \mathcal{C}} x_{t,p,c} q_{p,c} - \sum_{p \in \mathcal{S}_{out}} \sum_{c \in \mathcal{C}} x_{t,p,c} q_{p,c} + vol_{s,t-1} - vol_{s,t} = d_{t,s} \Delta t \quad \forall t \in T, \quad \forall s \in \Phi \quad (13)$$

$$\underline{vol}_{s,t} \leq vol_{s,t} \leq \bar{vol}_{s,t} \quad \forall t \in T, \quad \forall s \in \Phi \quad (14)$$

$$vol_{s,0} \leq vol_{s,T} \quad (15)$$

$$\sum_{c \in \mathcal{C}} x_{t,p,c} \leq 1 \quad \forall t \in T, \quad \forall p \in \mathcal{P} \quad (16)$$

$$x_{t,p,c} \in \{0, 1\} \quad (17)$$

Table 1
Example of pump combination.

Combination	Pump 1	Pump 2	...	Pump Ψ
1	1	0	0	0
2	1	1	0	0
⋮	⋮	⋮	⋮	⋮

The coupling of the two systems is through the pumping power terms, which are required to have the same values in the power balance constraints and in the WDS objective functions. Further details on the systems coupling based on Eq. (18) can be found in Appendix B.

$$A \bullet B \bullet x_{t,p,c} = P_{t,t}^p \quad \forall t \in T \quad (18)$$

2.4. Decoupled optimization

The decoupled optimization reflects individual decision-making with no communication between the two systems' operators. The WDS makes decisions based on the first objective function of minimal operational costs, where the PDS is bound to satisfy the power required by the WDS pumps.

$$x_{PDS}^* = \text{argmin}(1)$$

s.t. :

$$(2) - (10)$$

$$x_{WDS}^* = \text{argmin}(11)$$

s.t. :

$$(13) - (18) \quad (19)$$

2.5. Centralized coupled optimization

Here, all the decisions are made simultaneously by an omniscient decision maker. In this case, the decisions in both systems are made based on the emergency conditions. The WDS will adjust its operation to support LS minimization.

$$x_{PDS}^* = \text{argmin}(1)$$

s.t. :

$$\begin{aligned} (2) - (10) \\ (13) - (18) \end{aligned} \quad (20)$$

2.6. Coordinated optimization

The essence of the proposed method lies in the design of a communication protocol between the water and power utilities, aimed at optimizing a shared objective. The common objective is minimizing LS under emergency scenarios while the operational costs of the WDS are in lower priority and the power utility will be willing to pay the extra costs of the WDS. Ideally, a perfect knowledge program would be implemented, where all data and constraints from both systems are integrated into a single optimization problem. However, due to the independent management of water and power utilities in real-life systems, such comprehensive integration is impractical. Instead, we propose a methodology that accommodates limited information sharing to effectively enhance the common objective. Based on the bilevel optimization (BO) paradigm [44], the formulation includes a leader and a follower. Unlike the classic BO setup, the follower gets the opportunity to respond to the leader's policy and ask for adoptions that serve the common objective. The leader is the WDS which, unless received a notification from the PDS, optimizes pump scheduling to minimize costs. The follower is the PDS which adjusts the generation schedule to the WDS operation. During extreme scenarios such as when the PDS is disconnected from the main grid and LS becomes inevitable, the following steps are taken: First, the power operator solves an LS minimization problem while considering pumping loads as flexible loads. This step obtains an auxiliary solution that identifies the optimal pumping schedule from the PDS perspective. It is used solely as an auxiliary to support the next stages and will not be implemented. This is because it is assumed that in

emergency scenarios water demand is a critical load, so the WDS can adjust its operation to support the PDS objective of reducing LS, but without compromising the satisfaction of water demand. Therefore, when the PDS solves the first stage inner problem it ensures to allocate at least the nominal power that the WDS typically uses. This prevents the WDS from being limited in subsequent stages of the method due to the auxiliary solution. In other words, pumping loads are effectively assigned an infinite criticality index, which imposes a very high cost for LS affecting these loads. This prioritization ensures that pumping loads are treated as essential in the LS minimization process. It is noted that the first stage is only an auxiliary stage that supports the decision-making of the subsequent stages. In case the WDS will be able to satisfy its constraints with less power this will be reflected in the next stage results, in the adjusted WDS pumping schedule.

Next, the power operator generates a penalty matrix ω based on the optimal pump schedule, where $\omega = (1 - x_{WDS,1}^*)$, where $x_{WDS,1}^*$ is the optimized pump schedule from the power operator's perspective. The penalty matrix is delivered to the water operator which uses it to solve an inner "emergency" optimization problem with the objective function given in Eq. (12). The penalties act as the new objective coefficients in the WDS optimization, which directs the WDS to prioritize pumping from pumps that are connected to the most desired buses and at times that will serve best the PDS LS objective. This matrix of penalties is the only information exchange between the utilities.

Another adoption taken by the WDS is the relaxation of the final tank volume constraint in Eq. (15). Since this is an emergency scenario, tanks are allowed to not return to their initial volume. In this study, the required final volume was set to the min between the initial volume and two hours of average demand according to common emergency volume criteria [45]. The solution of the WDS inner resilience optimization is delivered as improved pump scheduling to the PDS to finalize the emergency response strategy. The process is depicted in Fig. 2.

2.7. Costs analysis

The response to PDS line outages, supported by operational adjustments within the WDS, entails certain costs for the water utility. While LS is minimized, the water utility incurs additional expenses due to alterations in the pumping schedule. To address this issue, it would be beneficial to monetarily quantify the additional costs carried out by the WDS due to the interventions for LS reductions. Furthermore, these costs are compared with the Cost of Energy Not Served (CENS). This analysis of the surplus cost of the water utility as a result of emergencies could also facilitate the calculation of potential subsidies between the two utilities, promoting a more collaborative financial relationship. The additional costs for the WDS are calculated as a function of the difference in pumping costs by the decoupled and coordinated approaches during the emergency and the costs associated with the missing volume of tanks at the end of the emergency, as outlined in Eq. (21). This equation consists of two components: the first calculates the cost difference during the emergency hours, reflecting the increased operational costs of the WDS; the second estimates the cost to replenish the missing water volume at the end of the emergency, factoring in the average specific power of the WDS pumps and applying the on-peak tariff, under the assumption that the replenishment occurs during on-peak hours.

$$\Delta_{WDS} = (Cost_{coordinated} - Cost_{decoupled}) + (Vol_{decoupled,T} - Vol_{coordinated,T}) \cdot \bar{SP} \cdot \tau_{on-peak} \quad (21)$$

Where $Cost_{coordinated}$ and $Cost_{decoupled}$ are calculated operational costs for the coordinated and decoupled strategies respectively. Similarly, $Vol_{coordinated}$ and $Vol_{decoupled}$ are the final volume of water tanks at the end of the emergency for each strategy. $\bar{SP}$ is the average specific power of

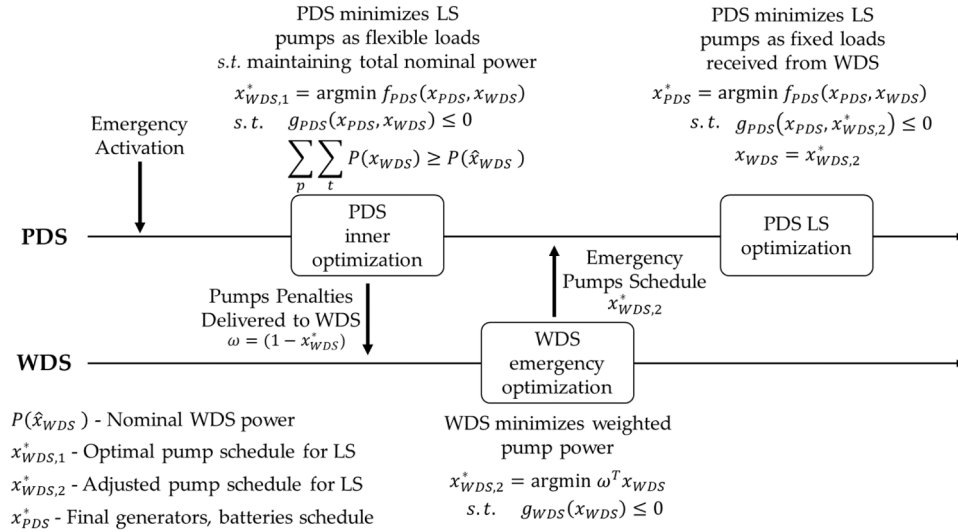


Fig. 2. Coordinated optimization communication order.

the WDS pumps and $\tau_{on-peak}$ is the on-peak electricity tariff.

The change in CENS is calculated based on the difference between the objective values of the decoupled and the coordinated approaches such that the difference in LS is the benefit of the coordinated approach. According to [46], the value of not-served energy (kWh) is a function of the outage duration. In the range of the duration considered in this study (6–24 hours) the price of not served kWh can be approximated by linear regression as shown in Eq. (22):

$$\Delta_{CENS} = (0.025 + 1.2 \cdot T) \cdot (LS_{decoupled} - LS_{coordinated}) \quad (22)$$

Where T is the emergency duration, and LS_i is the LS results from strategy i .

2.8. Random emergency scenarios

A random scenarios generator was developed to support the evaluation of the proposed methodology. The scenarios were generated by drawing sets of random parameters that characterize each scenario. These parameters include the start time of the scenario, its duration, scaling factors (multipliers) of the water demand, power loads, and PV availability profile. Additionally, the number of damaged power lines and the index of damaged (failed) lines. Lastly, the state of tanks and batteries at the start of the emergency, meaning the water volume and energy stored in tanks and batteries respectively. These parameters generate a range of scenarios different from each other in the level of severity. For example, in a scenario that occurs during a peak consumption day where water and power consumption are at their high range, it will be more difficult to reduce LS since the systems have fewer reserves. Similarly, some power lines are more critical than others, charged batteries are desired at the beginning of an emergency, etc. In this context, the amount of LS results from a scenario indicates its severity. All the parameters were drawn from a uniform distribution according to the configuration presented in Table 2.

3. Results

The proposed method was evaluated through a case study integrating a small WDS pumping water from three wells with a modified version of the 33-bus system benchmark. The case study is designed to simulate a local energy microgrid capable of off-grid operation, the microgrid voltage is 13.3 kV. By combining local renewable generators with storage devices and emergency diesel generators, microgrids can sustain off-grid operation for extended periods. This shift to microgrids

Table 2

Random emergency scenarios configuration.

Parameter	Distribution	Min	Max
Start time	Uniform	0	23
Duration	Uniform	6	24
Water demand factor	Uniform	0.8	1.2
Power load factor	Uniform	0.8	1.2
PV factor	Uniform	0.8	1.2
Number of outage lines	Uniform	1	2
Water tanks initial state	Uniform	0.2	1
Batteries initial state	Uniform	0.2	1

helps mitigate the instabilities in the main grid caused by the intermittent nature of renewable energy. Additionally, it ensures a reliable and high-quality power supply for consumers within these microgrids, who are shielded from the fluctuations and disruptions affecting the main grid [47].

Accordingly, the discussed microgrid is equipped with distributed energy resources, including PV and energy storage systems as suggested by [48]. To the modified 33-bus PDS we added a WDS that draws water from three local wells. The WDS contains two pressure zones such that the wells pump water into the lower pressure zone governed by tank t1 (3435 m³) and a booster pump lifts a portion of the water to the higher elevation consumers governed by tank t2 (1235 m³). The well pumps and the booster pumps both represent centrifugal pumps and are modeled the same in the optimization models. The wells are equipped with emergency diesel generators ensuring their functionality during emergencies. These generators can also provide power to the PDS. (Fig. 3)

To rigorously evaluate the ability of the proposed coordinated optimization ability to reduce LS it was tested against a range of 1000 emergency scenarios based on the distributions presented in Table 2.

For each random scenario, LS was minimized by all three operational strategies: decoupled, centralized coupled, and coordinated optimization (the reader is referred to the definitions in the methodology section). The effectiveness of the three strategies was statistically compared as presented in Table 3. The values in the table represent the amount of LS in kWh and the values within the parentheses show the deviation from the centralized coupled approach which constitutes a benchmark. The centralized coupled approach outperforms the other strategies epitomizing the ideal of perfect knowledge and coordination between WDS and PDS. The decoupled approach resulted in a 7.86 % increase in LS compared to the centralized coupled approach. Remarkably, the

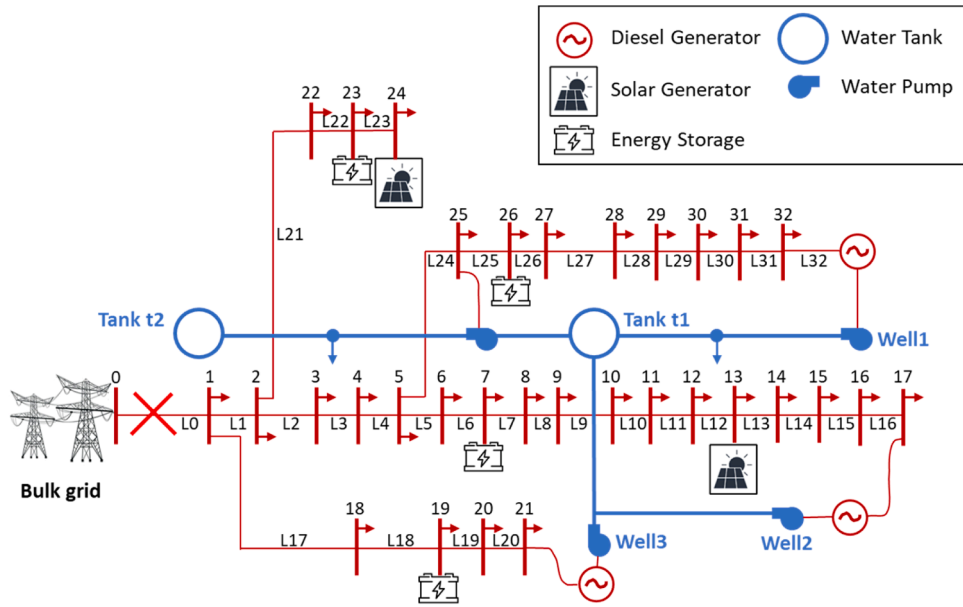


Fig. 3. Case study layout.

Table 3

Statistical comparison of total LS (kWh) between operational strategies.

	Centralized Coupled	Coordinated	Decoupled
Mean	5922.6	5979.4 (0.96 %)	6377.3 (7.86 %)
STD	3606.4	3644.0 (1.04 %)	3715.5 (3.03 %)

proposed coordinated optimization approach showcased its effectiveness by limiting the LS increase to a mere 0.96 % above the centralized coupled strategy. Furthermore, the standard deviation (STD) of the coordinated optimization solutions is smaller than the decoupled solutions indicating the consistency of the coordinated optimization supreme. The results demonstrate how the coordinated optimization strategy is a practical solution for utilizing WDS flexibility during an emergency.

Theoretically, the centralized coupled strategy reduces LS the most. However, it is impractical since it requires a single control center to operate both systems, which is rarely the case. The presented coordinated optimization approach performs nearly as effectively as the centralized one. Still, implementing it is much more feasible and requires only simple data transmission between the utilities.

Fig. 4 illustrates the potential of LS reduction across varying degrees of scenario severity. Where severity is estimated according to the LS results from the decoupled strategy (x-axis). The y-axis indicates the percentage reduction in LS achieved through the centralized coupled and coordinated strategies. As the scenario severity increases, the potential for LS reduction diminishes for both strategies. This observation aligns with expectations, as the WDS's contribution to the total PDS loads is finite, thus limiting its capacity to shift power post-emergency

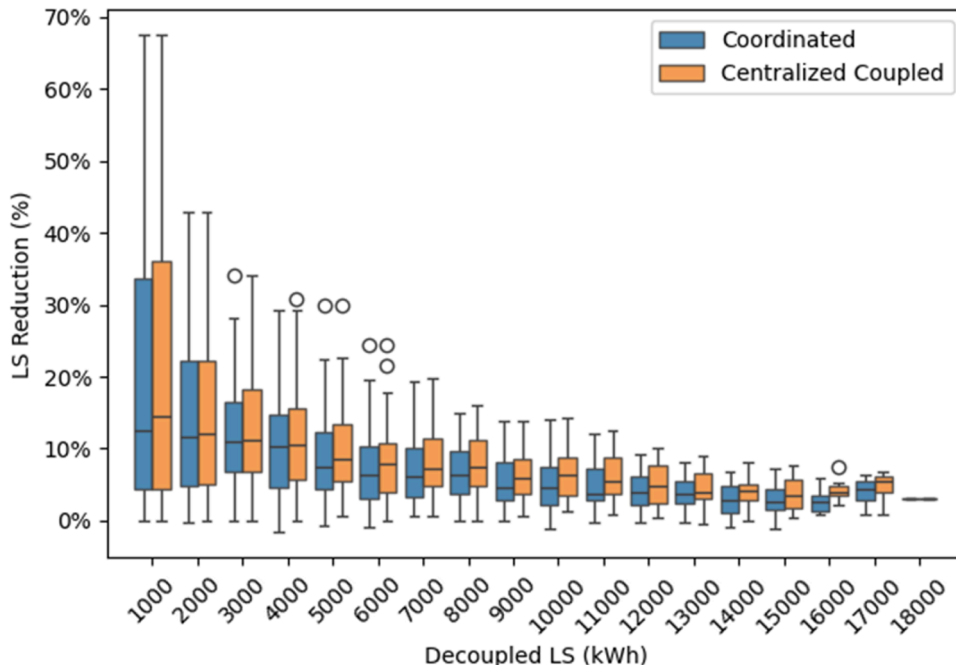


Fig. 4. LS reduction vs LS in a decoupled decision-making.

during severe emergencies. Notably, WDS can facilitate a reduction in LS ranging from almost 70 % in the milder scenarios to about 5 % in the most severe ones. Although the centralized coupled boxes present slightly larger LS reduction (boxes are slightly higher), the two approaches perform almost equivalently with similar LS reduction ranges. In 21 out of the 1000 random scenarios, the decoupled approach resulted in infeasible solutions, meaning that the WDS planned schedule is infeasible and might cause additional water outages alongside power disruptions. In contrast, the centralized coupled and coordinated approaches yielded feasible solutions in all these cases. Out of the three optimization strategies, the decoupled one is the most stringent thus presenting the greatest difficulty in finding a feasible solution. It is noted that the optimization problems are of non-linear mixed integer type. By relaxing the solver optimality gaps these infeasible solutions were eliminated.

The potential for LS reduction based on WDS adjustment depends on the emergency properties. Fig. 5 presents the percentage of LS reduction against the various parameters that define the emergency scenarios (as outlined in Table 2). The initial state of tanks and batteries is represented by the average values of volume and energy respectively. Each scenario is depicted as a line across the different coordinates representing the scenario properties, with line colors according to the LS reduction rate. The leftmost axis shows the percentage reduction in LS. The next axis represents the duration of the emergency scenario, indicating that shorter emergencies correlate with greater LS reduction. The effect of the emergency start time is also evident, events starting during peak consumption hours (morning to afternoon) have a lower reduction potential compared to scenarios that start at nighttime. While the correlation between the water demands factor and LS reduction is unclear, lower power load factors are correlated with higher LS reduction rates. Similarly, higher values of the PV availability factor also allow for greater LS reduction. To further explore the impact of water demand, power load, and PV availability factors, each was analyzed independently while keeping all other scenario properties constant. When isolated, the correlation between these factors and LS reduction becomes clearer as presented in Fig. S.5 in the supplementary materials. Unexpectedly, Fig. 5 shows that the system initial conditions of initial tank volumes and initial batteries SOC are not correlated with LS reduction. However, when the initial conditions are isolated, a strong correlation emerges. Since the initial conditions impact the different optimization approaches the same, the LS reduction (improvement) is not prominent in the analysis. In terms of absolute values, a direct relationship is

observed between higher initial capacity and lower LS. Interestingly, while higher initial batteries SOC consistently reduces LS, initial tank volumes above a certain threshold do not further reduce LS. The results of the initial conditions analysis are presented in Figs. S.6 and S.7 in the supplementary materials.

The expenses associated with different scenarios were analyzed to quantify the monetary benefit of the proposed method. The additional costs of the water utility are the result of pumping costs during and after the emergency. The emergency cost for the power utility is quantified based on the cost of energy not served (CENS) to manifest the damage for the power consumers. Fig. 6 graphically illustrates the relationship between the additional emergency costs incurred by each system. The x-axis displays CENS, and the y-axis indicates the additional costs borne by the water utility. It is evident that the reduction in CENS is significantly larger in magnitude compared to the additional costs incurred by the WDS. While the continuous operation of systems primarily depends on operational expenses alone, the loss of service incurs further consequences and entails higher costs than the cost of power generation. This is demonstrated in Fig. 6 which shows that the missing volume of water can be pumped post-emergency with the price of routine power generation while reducing the damage to consumers dramatically.

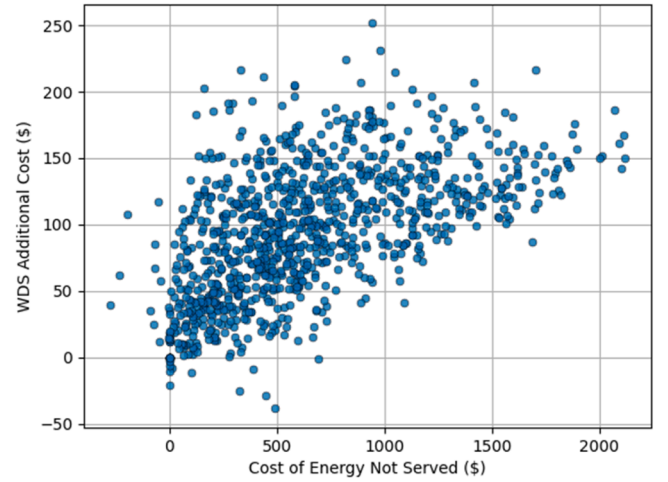


Fig. 6. WDS Additional pumping cost vs cost of energy not served.

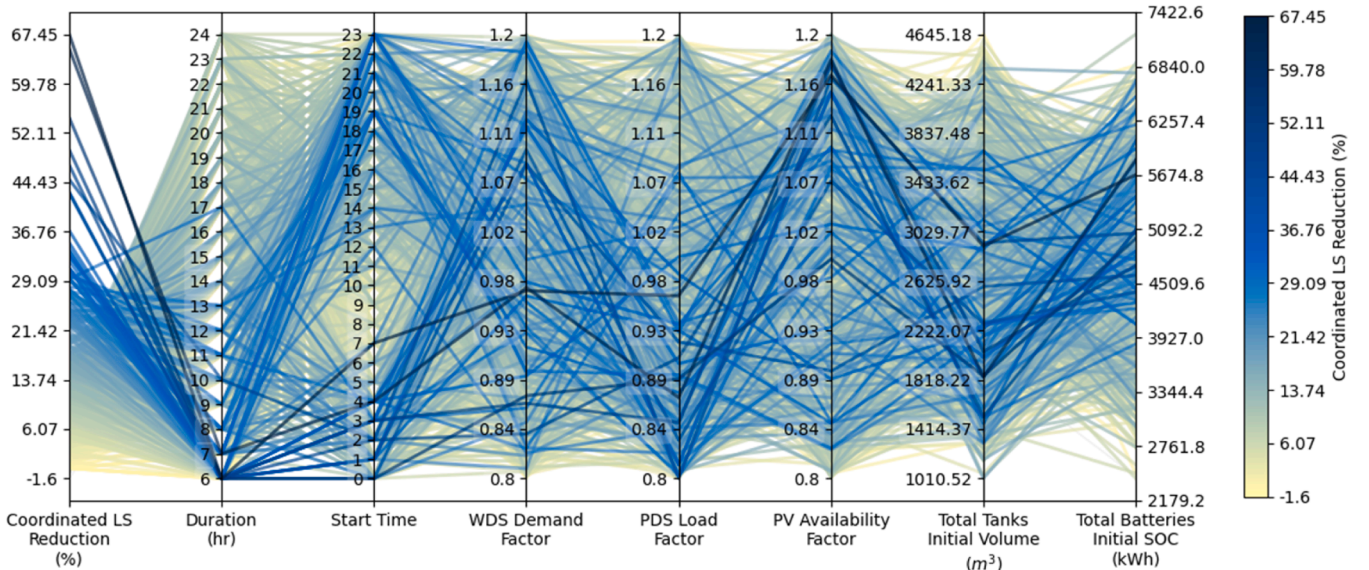


Fig. 5. LS reduction vs emergency scenarios parameters.

3.1. Edge scenarios analysis

Further analysis was conducted on specific scenarios to investigate the factors influencing the LS reduction capacity. Two extreme scenarios were chosen, the first one demonstrates a significant LS reduction whereas in the second one, only a minor reduction is achieved.

Scenario 1 – significant LS reduction

The scenario starts at 19:00 and lasts 12 hours. Lines 6 and 26 of the PDS are under outage. The tank volumes at the emergency start are 27 % and 46 % of their total capacity, and the batteries charging states are 55 %, 42 %, 66 %, and 24 %. For these conditions, LS will be 4199 kWh, 2944 kWh, and 2942 kWh for the decoupled, coordinated, and centralized coupled strategies respectively, exhibiting an almost 30 % reduction in LS. The WDS response to a scenario can be reviewed through the tank volume during the emergency as shown in Fig. 7, the pump schedule and power depicted in Figs. S.1 and S.2 in Appendix A. The emergency starts in the evening, right before the off-peak tariff period. Uninformed of the emergency event, the decoupled solution is such that the WDS pumps as much as possible during this time which can be seen in Fig. 7 where the tanks are constantly filled during the nighttime. With the PDS delivering the emergency signal, the WDS reduces pumping to the minimum possible without violating tank minimum volumes. This result indicates that emergencies during off-peak periods are more convenient. While from a pure PDS perspective, this might seem intuitive, the proposed methodology sheds light on how WDS has greater potential to support emergencies during off-peak periods.

Another factor impacting the results is the water demand and power load factors of 0.93 and 0.8, respectively. The lower consumption of water and load reduces the systems' strain allowing for a better supply-demand balance. The tank volumes are quite empty with 27 % in the main (lower) tank and 46 % in the upper (smaller) tank, and the battery states are around 50 % charged. These initial conditions are not favorable for the objective of LS reduction, yet this scenario is one where WDS reduces LS the most.

Scenario 2 – minor LS reduction

A second emergency scenario illustrates conditions where the WDS contribution to the LS reduction is limited. This emergency starts at 06:00 and lasts 19 hours. Lines 14 and 25 of the PDS are under outage. The tank volumes at the emergency start are 31 % and 53 % of their total capacity, and the batteries charging states are 83 %, 66 %, 74 %, and 35 %. For these conditions, LS will be 11474 kWh, 14414 kWh, and 14406 kWh for the decoupled, coordinated, and centralized coupled strategies respectively, indicating an LS reduction of 0.5 %. It is noted that both centralized coupled and coordinated approaches only minorly reduce LS, pointing out that the cause for the minor improvement is not

the method disadvantages but stems from the scenario itself which contains critical conditions that prevent LS reduction. Fig. 8 presents the tank volumes during the emergency, the pump schedules and the power depicted in Figs. S.3 and S.4 in Appendix A.

This emergency scenario falls in the daytime when power and water consumption are at their upper edge, so the systems have fewer redundancies to decrease LS. The water demand factor is 1.09 and the PDS loads factor is 1.11. These values indicate a peak consumption day, adding another challenge on top of the emergency scenario. The two water tanks start the emergency with 31 % and 53 % of their capacity. This exceeds WDS operation standards where pumps are expected to operate during the night and start the morning demand cycle when full. This exception from common practical standards is one of the causes that prevent the WDS from shifting pumping and enable further LS reduction. Lastly, the PV availability is at its low edge, with a PV factor of 0.8 which further limits the power generation capacity.

4. Conclusions

The integrated management of WDS and PDS was investigated through a coordinated optimization framework, compared with decoupled and centralized coupled strategies aimed at minimizing LS during emergencies. Previous studies on joint multi-system optimization often assumed full transparency and cooperation between operators, which is unrealistic. Alternatively, some studies suggested independent frameworks where each utility optimizes its objectives independently, but this is typically sub-optimal. The current study proposes a middle path of a coordinated optimization approach. This approach is based on practical communication between utilities, leading to better overall performance. The coordinated strategy proved highly effective in managing emergency conditions for the examined case study of a local microgrid connected to a small WDS, consistently reducing LS compared to decoupled decision-making across a range of simulated emergency scenarios. For this case study, the coordinated strategy's performance closely approached the optimal benchmark results of the centralized coupled strategy. While the proposed method does not rely on assumptions that limit its applicability to the specific case study, its effectiveness was demonstrated only on a single configuration, with variations primarily in the simulated scenarios. Therefore, future research should focus on implementing and validating the method across different categories of networks to further confirm its generalizability and robustness. Additionally, the study analyzes the factors that impact the severity of emergency scenarios and the effectiveness of the proposed method in reducing LS. It was found that WDS operational adjustments are more effective in scenarios where some redundancy is available, such as during nighttime scenarios. Future research should analyze different

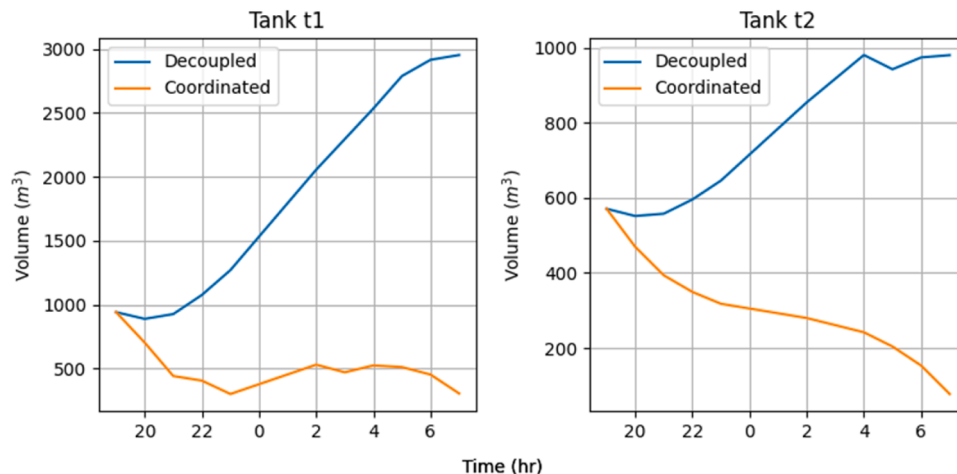


Fig. 7. WDS tanks behavior for significant LS reduction scenario.

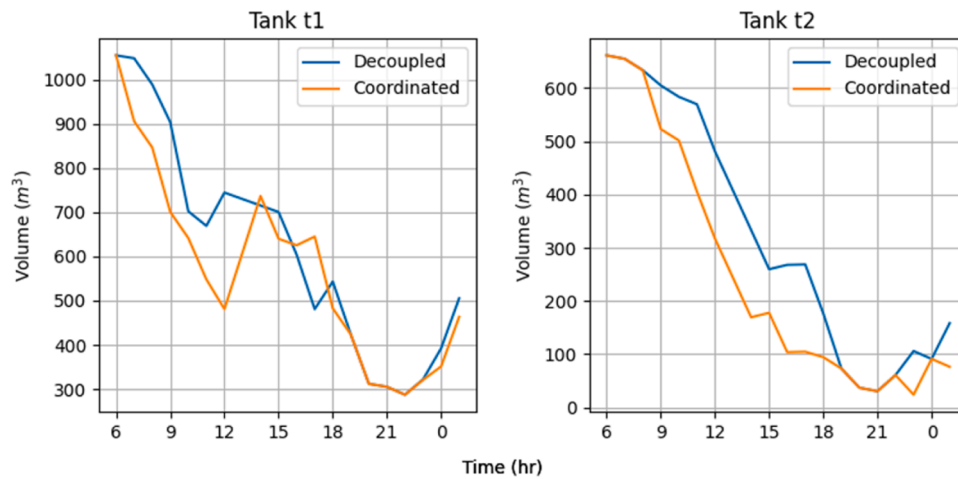


Fig. 8. WDS tanks behavior for minor LS reduction scenario.

configurations of interconnected PDS and WDS to further assess the generality of the method. Another potential future research is the routine operation of WDS and its effect on the resilience of both systems, for example, by maintaining higher minimum tank levels. Moreover, future research is needed to further explore this decision space from a stochastic perspective, accounting for the uncertainty inherent in emergency scenarios. In conclusion, this study contributes to the ongoing discourse on integrated WDS-PDS management by demonstrating that limited communication between PDS and WDS is sufficient to significantly enhance the efficiency and resilience of these critical infrastructure systems.

CRedit authorship contribution statement

Panteli Mathaios: Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Eliades Demetrios:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Vrachimis Stelios:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization. **Ostfeld Avi:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Shmaya Tomer:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization. **Navon Aviad:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization. **Perelman Gal:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Avi Ostfeld reports financial support was provided by The Bernard M. Gordon Center for Systems Engineering at the Technion. Demetrios G. Eliades reports financial support was provided by The ERDF and the Rep. of Cyprus. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.segan.2025.101643](https://doi.org/10.1016/j.segan.2025.101643).

Data availability

The data and code used in this research are available on the following GitHub repository: <https://github.com/GalPerelman/pds-wds-coordinated>

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